

Does the certainty of observation depend on perspective?

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Abstract

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1 Operator Talk

You may now be familiar with the notion of creation and annihilation operators in quantum mechanics (as discussed in previous documents), for examples, when we promote a classical scalar field:

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^* e^{ipx}) \quad (1)$$

to an operator, we have:

$$\hat{\phi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{a}_p e^{-ipx} + \hat{a}_p^\dagger e^{ipx}). \quad (2)$$

We can see that both the creation and annihilation operators correspond to positive and negative frequency solutions to the classical equation of motion. In curved spacetime, there isn't exactly a globally defined time direction, so globally defining the idea of creation and annihilation operators is not possible. For this reason, we can express local operators in terms of the operators defined at some other point in spacetime. To do this, we use the *Bogoliubov transformation*:

$$\hat{b} = C\hat{a} + D\hat{a}^\dagger \quad (3)$$

$$\hat{b}^\dagger = C^*\hat{a} + D^*\hat{a}^\dagger. \quad (4)$$

We require that $\hat{b}$ and $\hat{b}^\dagger$ satisfy the same commutation relations as $\hat{a}$ and $\hat{a}^\dagger$:

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (5)$$

Since the Bogoliubov transformation is canonical, this will be the case, but we need to find some restrictions on C and D .

$$[\hat{b}, \hat{b}^\dagger] = (C\hat{a} + D\hat{a}^\dagger)(C^*\hat{a} + D^*\hat{a}^\dagger) - (C^*\hat{a} + D^*\hat{a}^\dagger)(C\hat{a} + D\hat{a}^\dagger) \quad (6)$$

This simplifies to:

$$[\hat{b}, \hat{b}^\dagger] = (|C|^2 - |D|^2) [\hat{a}, \hat{a}^\dagger] \quad (7)$$

Which implies:

$$|C|^2 - |D|^2 = 1 \quad (8)$$

2 A Peculiar Situation

Suppose we wanted to consider the expectation value of the number of particles for some state $|n\rangle$:

$$\langle n | \hat{b}^\dagger \hat{b} | n \rangle = \langle n | \hat{N} | n \rangle \quad (9)$$

We find the number operator for b to be:

$$\hat{b}^\dagger \hat{b} = (C^*\hat{a} + D^*\hat{a}^\dagger)(C\hat{a} + D\hat{a}^\dagger) \quad (10)$$

This simplifies to:

$$\hat{b}^\dagger \hat{b} = (|C|^2 + |D|^2)\hat{a}^\dagger \hat{a} + C^*D(\hat{a}^\dagger)^2 + D^*C(\hat{a})^2 + |D|^2 \quad (11)$$

Taking the expectation value for each of the operators in these terms, we have:

$$\langle n | \hat{a}^\dagger \hat{a} | n \rangle = n \quad (12)$$

$$\langle n | (\hat{a}^\dagger)^2 | n \rangle \propto \langle n | n + 2 \rangle = 0 \quad (13)$$

$$\langle n | (\hat{a})^2 | n \rangle \propto \langle n | n - 2 \rangle = 0 \quad (14)$$

Therefore, (11) further simplifies to:

$$\langle n | \hat{b}^\dagger \hat{b} | n \rangle = (|C|^2 + |D|^2)n + |D|^2 \quad (15)$$

Now suppose we had an observer, A, falling towards the event horizon of a black hole. They make observations of photons associated with the a operators in A. Suppose they observe no photons, that is:

$$\langle 0 | \hat{a}^\dagger \hat{a} | 0 \rangle = n = 0 \quad (16)$$

Now suppose we have an observer, B, in an approximately flat region of space-time, observing the same system as A. The operators in this case are transformed to b operators. The number of particles observed in B's frame is:

$$\langle 0 | \hat{b}^\dagger \hat{b} | 0 \rangle = (|C|^2 + |D|^2)(0) + |D|^2 = |D|^2 \quad (17)$$

So interestingly, observer B sees a non-zero number of particles where observer A sees none. The observed number of particles is then dependent on perspective.